

SAI TEJASWAR REDDY DALLI

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PROFESSIONAL SUMMARY

AI/ML Engineer with around 5 years of experience building scalable machine learning, generative AI, and data-driven solutions across financial services and enterprise domains. Skilled in Python, TensorFlow, PyTorch, XGBoost, LLMs, RAG, LangChain, AWS, and GCP for developing predictive models, fraud detection systems, NLP applications, and AI automation platforms. Experienced in MLOps, Docker, Kubernetes, MLflow, CI/CD pipelines, SQL, Tableau, and Power BI to deliver scalable, high-performance AI solutions and business insights.

WORK EXPERIENCE

American Express, USA | *AI Engineer*

August 2025 – Present

- Built enterprise-grade AI solutions using GPT-4, LangChain, and RAG frameworks with Pinecone and ChromaDB vector databases, enabling real-time retrieval of financial knowledge across 15M+ customer interaction records and reducing analyst research time by 45%.
- Engineered fine-tuned transformer models using TensorFlow, Python, and LlamaIndex to automate fraud risk classification and intelligent customer support workflows, improving prediction accuracy by 28% and reducing manual case handling efforts by 35%.
- Developed scalable MLOps infrastructure with MLflow, GitHub Actions CI/CD pipelines, and automated model monitoring, accelerating deployment cycles by 40% while ensuring governance, reproducibility, and compliance across AI/ML production environments.
- Implemented explainable AI frameworks using SHAP and LIME for credit risk and underwriting models built on XGBoost algorithms, increasing model transparency for regulatory audits and improving stakeholder trust in AI-driven decision-making processes.
- Spearheaded development of multi-agent AI systems leveraging LangChain orchestration and RAG pipelines to automate financial operations, customer query resolution, and internal knowledge retrieval, increasing operational efficiency by 30% and reducing response turnaround times by 50%.

Infosys Pvt Ltd, India | *AI/ML Engineer*

February 2020 – January 2023

- Designed and deployed fraud detection pipelines using BigQuery ML, GCP Dataflow, and PostgreSQL to process over 20M+ transactional records daily, improving fraud identification accuracy by 32% and reducing false-positive rates by 25%.
- Built scalable ML workflows with TensorFlow Extended (TFX), Vertex AI, Docker, and Google Kubernetes Engine (GKE), streamlining model training, validation, and deployment processes while reducing production release cycles by 40%.
- Optimized predictive analytics and forecasting models using LightGBM, ARIMA, and Prophet for financial and operational datasets, increasing forecasting accuracy by 27% and enabling proactive business planning across enterprise functions.
- Implemented computer vision and NLP solutions using Mask R-CNN, Named Entity Recognition (NER), and NLTK to automate document classification, invoice data extraction, and entity recognition workflows, reducing manual processing efforts by 50%.
- Instituted MLOps best practices for CI/CD automation, model monitoring, and performance governance while developing Tableau dashboards for model KPIs and operational insights, improving stakeholder reporting efficiency and reducing issue resolution time by 35%.

Hexaware Technologies, India | *Junior Data Scientist*

January 2019 – January 2020

- Analyzed structured and unstructured enterprise datasets using Python, SQL, Pandas, NumPy, and exploratory data analysis (EDA) techniques to identify operational trends and improve reporting accuracy by 25% across client delivery projects.
- Developed classification and predictive models using Decision Trees, Random Forests, Naive Bayes, XGBoost, and Scikit-learn, increasing customer churn prediction accuracy by 30% and supporting data-driven business recommendations for enterprise clients.
- Engineered clustering and dimensionality reduction solutions using K-Means, DBSCAN, and Principal Component Analysis (PCA) to segment customer behavior patterns and optimize marketing campaign targeting, improving campaign conversion rates by 18%.
- Built scalable data processing workflows using PySpark, MongoDB, and AWS services, including SageMaker, S3, and Lambda, to automate data ingestion and model deployment pipelines, reducing manual processing time by 40%.
- Created NLP-driven analytics solutions using NLTK and PyTorch for sentiment analysis and text classification while designing Power BI dashboards for KPI tracking and operational insights, improving stakeholder reporting efficiency and reducing turnaround time by 35%.

PROJECTS

LLM Fine-Tuning & Evaluation Pipeline

- Fine-tuned pre-trained LLaMA large language model for domain-specific text generation tasks using custom training pipelines.
- Built a model evaluation pipeline to assess fine-tuned model performance and validate output quality.
- Documented end-to-end LLM fine-tuning and deployment workflow, including model configuration and inference setup

Ficopilot - AI Financial Agent

- Built an AI-powered financial analysis application using Streamlit that enables natural language queries on uploaded Excel financial data.
- Implemented automated analysis of revenue vs budget variance, gross margin trends, operating expenses, EBITDA, and cash runway metrics, improving financial visibility and reducing manual reporting efforts.
- Created interactive Plotly visualizations for financial insights with data validation and error handling

SKILLS

Programming Languages & IDEs: Python, R, SQL, Jupyter Notebook, Google Colab, VS Code, PyCharm

Machine Learning & Deep Learning: Decision Trees, Random Forests, Naive Bayes, K-Means, DBSCAN, Principal Component Analysis (PCA), LightGBM, Mask R-CNN, XGBoost, SHAP, LIME, ARIMA, Prophet, Hugging Face Transformers

Deep Learning Frameworks: TensorFlow, Keras, PyTorch

AI & Generative AI Technologies: LangChain, LangGraph, LlamaIndex, Retrieval-Augmented Generation (RAG), LLaMA, GPT-4, Claude, Fine-tuning (LoRA, QLoRA, PEFT), Prompt Engineering, CrewAI, Multi-Agent Systems

Natural Language Processing (NLP): BERT, RoBERTa, Named Entity Recognition (NER), NLTK

MLOps & DevOps: MLflow, Kubeflow, TFX, Docker, Kubernetes (GKE, EKS), Terraform, CI/CD, GitHub Actions, Jenkins, Model Deployment, Model Monitoring

Cloud Platforms: Google Cloud Platform (Vertex AI, BigQuery, Dataflow, AI Platform, Cloud Composer), AWS (SageMaker, S3, Lambda), Azure (ML Studio, Databricks)

Libraries & Tools: Scikit-learn, Pandas, NumPy, SciPy, PySpark, Matplotlib, Seaborn, OpenCV

Data Visualization & Databases: Tableau, Power BI, SQL Server, PostgreSQL, MongoDB, Vector Databases (Pinecone, FAISS, ChromaDB)

EDUCATION

Master of Science in Computer Science

May 2025

California State University, Dominguez Hills, California, USA

GPA: 3.52

Bachelor of Technology in Computer Science

May 2019

Raghu Engineering College, Andhra Pradesh, India

GPA: 3.0

CERTIFICATIONS

AWS Certified AI Practitioner | Generative AI with AWS – Udacity | Model Context Protocol (MCP) for Beginners by Microsoft | Scientific Computing with Python | AWS Academy Graduate -- Machine Learning through Application | Infrastructure & Application Modernization with Google Cloud